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Radio-Quiet Ionosonde Observations for Monitoring Absorption in the HF Band

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Key Points:

- Ionosondes can observe without transmitting, recording signals of opportunity
- These radio-quiet observations can be used to monitor ionospheric absorption of HF radio
- Effects of absorption enhancements due to solar flares or particle events can clearly be identified

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Abstract This work presents a technique to monitor ionospheric absorption in the HF radio band using ionosondes. For this purpose, the ionosondes were run in a radio-silent mode, recording not reflections of their own emitted signals but oblique signals from transmitters of opportunity. We present first the quiet-time data, showing what the typical diurnal variations in the spectrograms are. We then demonstrate that enhancements of ionospheric absorption resulting from either solar X-rays or particle precipitation events can be detected in these data. We show several examples of such signatures in the spectrograms, in time series integrated over a wide band of frequencies, and in small frequency band time series. The aim of this paper is to introduce this type of observations and demonstrate the principle of its use for absorption monitoring.

1. Introduction

Absorption of HF radio signals in the ionosphere primarily occurs in the D-region, the lower part of the ionosphere. Hence, absorption is strongly correlated to solar flares and particle precipitation events, which enhance D-region ionisation. The D-region of the ionosphere is commonly studied using RIOMETERS (Barta et al., 2024; Fiori et al., 2022; Little, 1954) or VLF beacons (e.g., Barta et al., 2022; Mitra, 1974), but there is an interest in additional data sources to provide more details and to fill gaps in observational coverage (e.g., Kauristie et al., 2021).

Various techniques are being developed to monitor the level of HF-band absorption and the ionisation in the D-region using existing ionosondes. One of the most widely used approaches is based on the minimum observable frequency on ionograms, commonly referred to as $f_{\min}$. This parameter corresponds to the lowest frequency at which reflected echoes can be detected and is sensitive to increases in non-deviative absorption in the lower ionosphere (Davies, 1990; Rishbeth & Garriot, 1969). During solar flares, increased ionisation in the *D* and lower *E* regions leads to stronger attenuation of low-frequency radio waves, resulting in higher $f_{\min}$ values and, in some cases, the temporary disappearance of low-frequency echoes. Owing to its long availability in ionogram records, $f_{\min}$ has been extensively used as a qualitative indicator of flare-induced absorption over a wide range of geographic regions (e.g., Barta et al., 2019; Denardini et al., 2016; Erdey et al., 2026; Sripathi et al., 2013; Tao et al., 2020).

Absorption signatures can also be identified through changes in the characteristics of the received echoes. One such method analyses the signal-to-noise ratio derived from ionograms (Curto et al., 2018; de Paula et al., 2022). In this method, the strength of the reflected signal is compared with the background noise level, and absorption events are identified by a significant degradation of the SNR relative to a quiet-time reference pattern representative of undisturbed conditions. This technique provides a time-resolved measure of signal attenuation during solar flare events.

A related technique focuses on variations in the amplitudes of ionosonde echoes. By analysing relative changes in received signal strength at selected frequencies, this amplitude-based technique enables the assessment of absorption affecting both the E- and D-regions of the ionosphere (Buzás et al., 2023). Large amplitude variations have been reported during disturbed conditions, demonstrating the sensitivity of this method to enhanced ionisation. Nevertheless, its applicability is limited in situations where strong ionospheric disturbances, such as intense geomagnetic storms or major solar flares, lead to severe signal fading or the complete absence of detectable echoes within the analysed frequency range.

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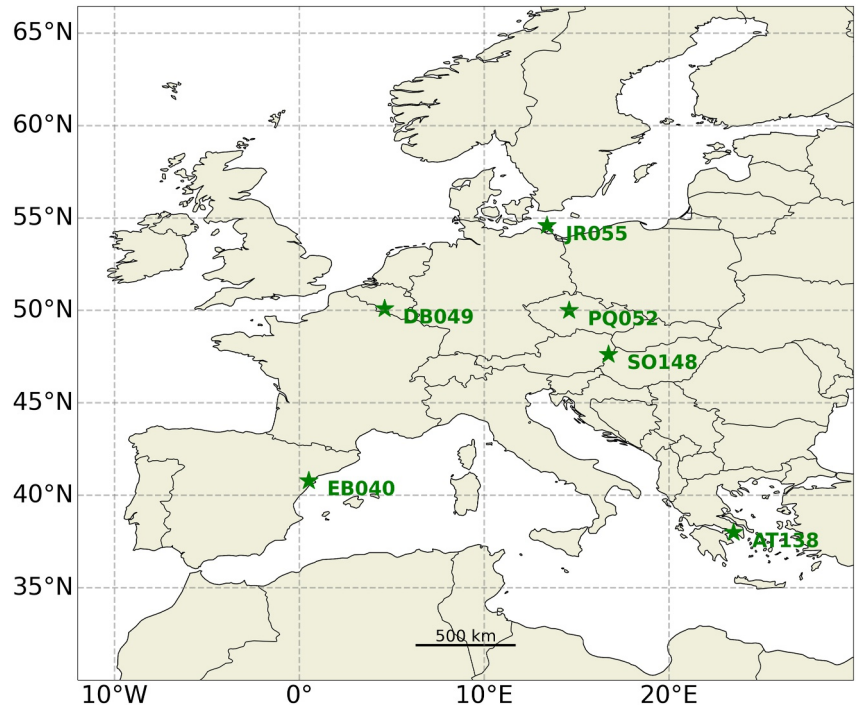


Figure 1. Location of DPS4D ionosondes used in this work.

The current paper presents an alternative approach, using ionosondes as passive receivers monitoring the strength of existing HF transmitters. This can, in principle, be done in two different ways, either by tuning to specific HF beacons of known strength–for example those of the International Beacon Project (Frissell et al., 2014)–or by monitoring the total noise received from any transmitter over the entire band. We employ here the latter method, recording all signals received in the HF band. This method allows direct monitoring of signal strength at various frequencies, including above the local f_oF_2, complementing the information that can be obtained using the various other ionosonde-based techniques mentioned above or other instruments.

This principle of measurements has been used by other ionosonde operators in the past, for example, in Tsai et al. (2014), but for the purpose of evaluating the noise level for the ionogram observations. The purpose here is instead to actively use this noise to monitor the lower ionosphere, as has been done by other researchers using SuperDARN noise measurements (Berngardt et al., 2018; Bland et al., 2018).

2. Observations and Data Processing

We use data from various DPS4D ionosondes (https://digisonde.com/digisonde.html) installed throughout Europe. Figure 1 shows the location of the ionosondes in Europe making radio quiet observations such as are analysed in this work; Table 1 provides the coordinates of the observatories. The measurements used here are

Table 1
European Ionosonde Observatories Making Radio-Quiet Measurements

URSI code	Observatory name	Latitude	Longitude
JR055	Juliusruh	54.6°	13.4°
PQ052	Průhonice	50.0°	14.6°
DB049	Dourbes	50.1°	4.6°
SO148	Sopron	47.6°	16.7°
EB040	Ebre	40.8°	0.5°
AT138	Athens	38.0°	23.5°

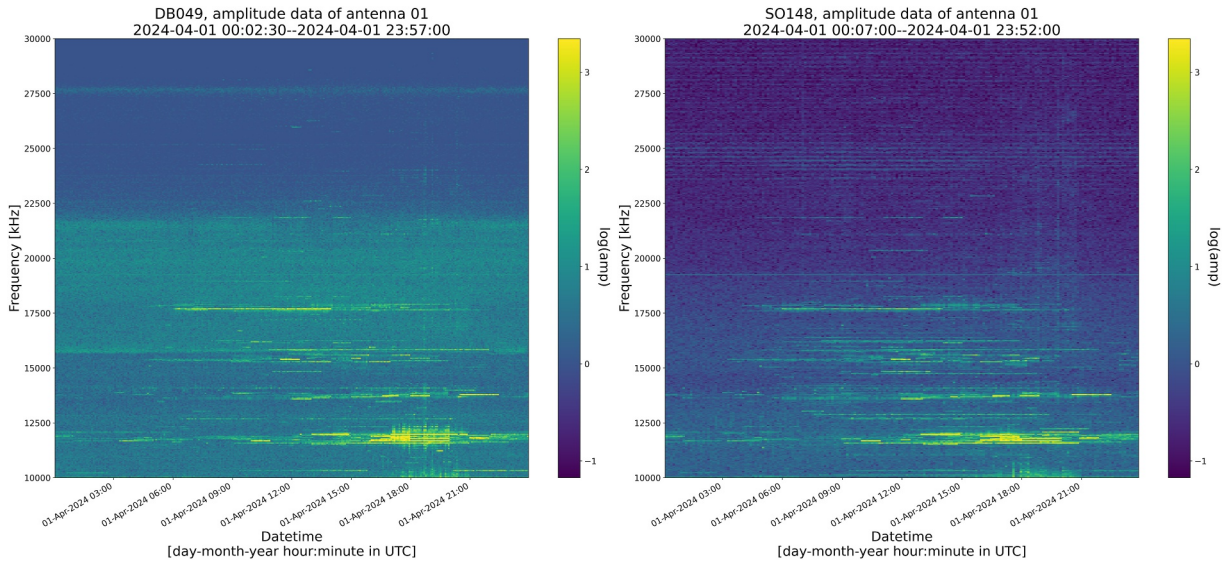


Figure 2. Example 24-hr spectrograms on the same day (2024-04-01) at the observatories of Dourbes (DB049, left) and Sopron (SO148, right).

similar to the traditional operation of an ionosonde, sweeping to all the frequencies of interest, except that the instrument is working in a receiver-only mode without transmitting any pulses.

Observations were started in February 2024, initially requested in support of the ROLSES experiment on the IM-1 *Odysseus* lunar lander (Burns et al., 2021; Hibbard et al., 2026). The data requested for that purpose was coverage of the 10–30 MHz range at 7.5-min cadence. Starting from September 2025, data was also collected at lower frequencies, down to 2.5 MHz, at some observatories (DB049, SO148, EB040).

The DPS4D employs an array of four receiver antennas, which are normally used as an interferometer to detect the direction of received signals, each consisting of two crossed loops, to allow detection of different polarisations. For the current work, the instrument was configured to receive O-polarised signals only. A simple time average of the signal at each frequency was recorded for each of the receiver antennas. The details of the instrument configuration are given in Appendix A.

Only data from a single antenna was used in this work. We verified that differences between the antennas at most observatories are insignificant, and in those observatories where some antennas are more affected by local noise than others, the least affected antenna was used. Data are sent to the Global Ionospheric Radio Observatory (Reinisch & Galkin, 2011), where they can be freely accessed as described in Appendix B.

Figure 2 shows a typical example of the full range (10–30 MHz) spectrograms for a full day (1 April 2024). The two panels represent two different observatories, DB049 on the left and SO148 on the right; the colour scale used for the received amplitudes is the same in both panels. The level of received noise depends on the receiver sensitivity and the local noise environment around each observatory. For example, the enhancement seen at DB049 around 28 MHz, which is continuous throughout the day and not seen at SO184, is clearly due to some local source close to the receiver antennas at DB049. Nevertheless, some features of the spectrogram can be seen to be very similar at all observatories at the same times.

3. Quiet Time Spectrograms

3.1. Median Spectrograms

As is evident from the spectrograms shown in Figure 2, different frequencies show maxima in received amplitudes at different times of day. The effects of absorption events on the received noise amplitudes are therefore best detected when comparing the received amplitudes to a median diurnal curve, with medians calculated for each frequency, or for small frequency bands, separately. When analysing any particular event, we calculate the median spectrogram from the data on the 7 days preceding and 7 days following the day of the event. The effects of solar flares are generally short, of the order of a few hours for strong flares, and flares occurring during the

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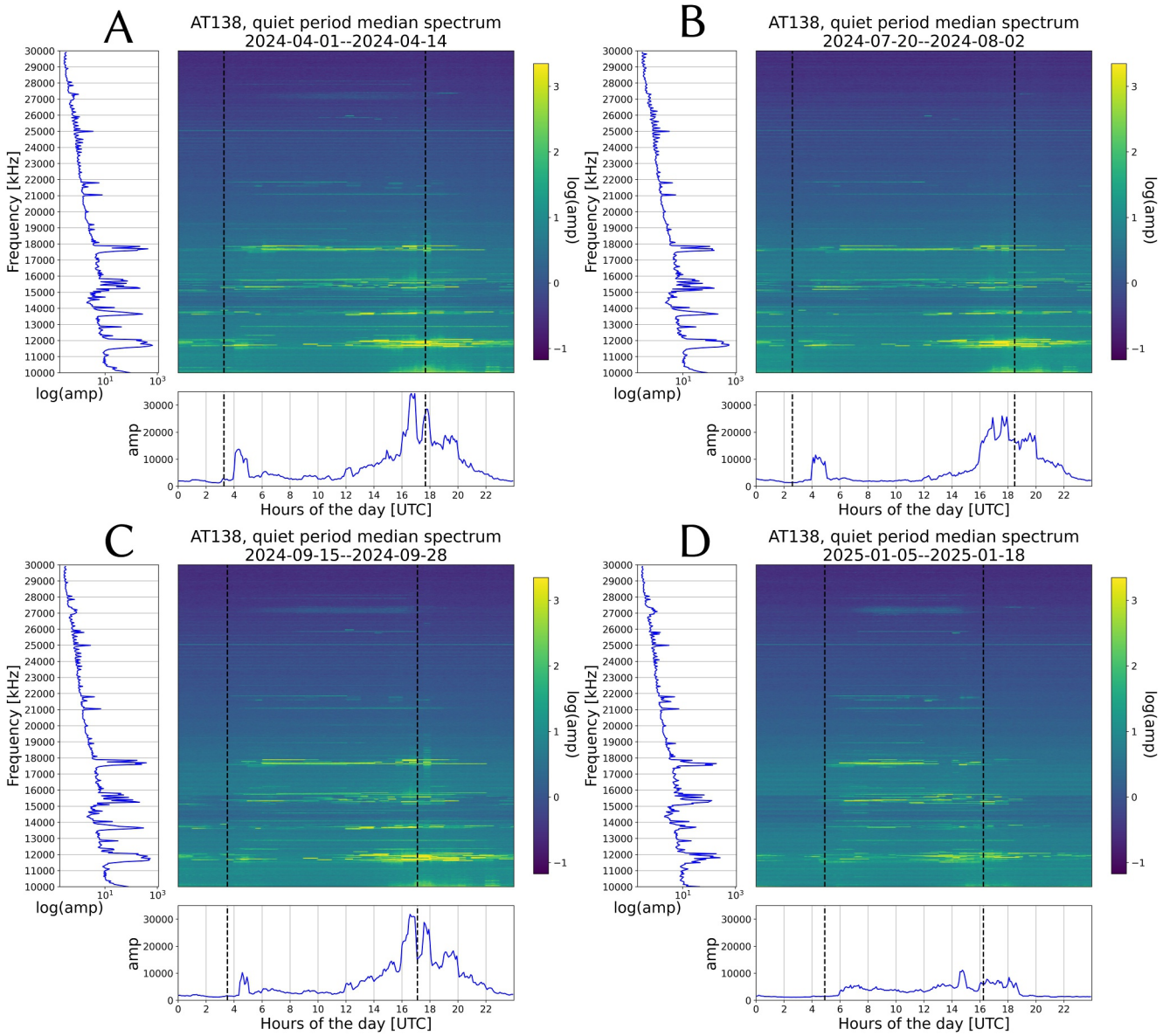


Figure 3. Quiet-time median spectrograms obtained from the Athens observatory. The vertical lines show the times of sunrise and sunset at 80 km altitude for the middle of the covered period.

reference days are therefore not expected to have a significant effect on the median spectrograms (Bland et al., 2018). Polar cap absorption events due to solar energetic protons on the other hand can last several days. Thus, especially when multiple particle-induced events occur in quick succession, a different period may have to be chosen to establish the quiet day curve. Such cases are not discussed in the current work, so for the examples included here we can rely on the medians calculated from these 14 days.

Figure 3 shows the median spectrograms obtained from the Athens ionosonde (AT138) for different 14-day periods representative for the different seasons. Below each spectrogram, a 24-hr time series of the median integrated over all frequencies is shown. On the left of the spectrogram, a time-integrated spectrum is shown.

The time-integrated spectra show a combination of two parts: a background noise present at all frequencies, with peaks in specific, relatively narrow bands superimposed on the background. The background shows a clear power-law dependence, with higher noise levels at lower frequencies. The wide-band noise is a combination of extraterrestrial noise and sky-wave propagation of noise generated primarily by lightning strikes (Bland

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et al., 2018, 2022). The narrow superimposed bands correspond to the frequency bands allocated for broadcasting applications. Within the considered range from 10 to 30 MHz, these bands are 11.60–12.10 MHz, 13.57–13.87 MHz, 15.10–15.80 MHz, 17.48–17.80 MHz, 18.90–19.02 MHz, 21.45–21.85 MHz, and 25.67–26.175 MHz.

It can be seen from the spectrograms that the largest variations over time occur in these broadcasting bands. This is understandable, as most of the signals received in these bands will be propagating from distance transmitters via the ionosphere. On the other hand, the power law background noise is likely due to local noise sources close to the observatory and to instrumental noise. This portion of the noise is not affected by ionospheric variability, and therefore not of much interest for the purpose of this paper.

For the different broadcasting bands, typical diurnal patterns can be observed from the spectrograms. The lower frequencies show increased amplitude of reception for a longer part of the day than do the higher frequencies, the latter only being enhanced closer to local noon. This is due to the typical diurnal pattern of maximum usable frequency, which facilitates reception of higher frequencies at the same distance when the ionisation of the *F* layer is the strongest. Usually, these diurnal patterns are interpreted as allowing more frequencies to be received at a fixed distance. For understanding the patterns seen in the spectrograms here, it is more convenient to consider the opposite interpretation: increased *F*-layer ionisation allows propagation of any particular frequency from a larger area around the observatory, thus explaining the observed patterns as changes in the integration area from which transmissions will be received.

For the lower frequencies considered, a clear decrease is seen close to local noon. For example, in all panels of Figure 3, the lowest broadcasting band (11.60–12.10 MHz) can be seen to exhibit lower amplitudes between about 06:00 and 12:00 UTC than earlier in the morning or later in the evening. This is the result of the increased ionisation in the *E* and *D* layers. At times when a frequency is being reflected from the *E* layer rather than the *F* layer, the area from which the frequency is being received is reduced due to the lower altitude of the reflection points. During the day, also the *D* layer will be present, leading to absorption of the lower frequencies.

The diurnal variations of the *E* and *F* layers vary throughout the year, and on a longer time-scale also with the solar cycle. However, in quiet times their variations over the 14-days period used to calculate the median is limited. Thus, while the median reference curves have to be calculated for each event individually, they should capture all relevant variations in the quiet-time *E* and *F* layers. One possible complication is presented by sporadic *E* layers, which do vary significantly from one day to the next. However, in most cases strong E_s layers are also not very long lasting and their effects on the median curves, similar to those of solar flares, can be assumed to be limited.

The panels below the spectrograms in subfigures A to D of Figure 3 show the diurnal, frequency-integrated pattern for the different seasons. The diurnal variations in the ionisation in the *F* and *E* regions lead to changes in the area from which transmissions can be received, while the *D*-layer variations result in changes to the level of absorption. The combined effect of these different factors results in various features in the diurnal patterns. Around sunrise and especially sunset the largest increases are seen in the received amplitudes. This greyline enhancement of ionospheric radio propagation is a well-known phenomenon (e.g., Lee, 1991; Lo et al., 2022). Notice that the pattern seen in panel D, representing January 2025, is markedly different from the others. This is due to a combination of seasonal effects as well as the somewhat higher solar activity in 2025 (measurements are not available for January 2024).

3.2. Extension to Lower Frequencies

Starting from 21 September 2025, some of the observatories (DB049, EB040 and SO148) lowered the lower frequency limit to 2.5 MHz (from the 10 MHz lower frequency limit used earlier and still at other observatories). Figure 4 shows an example of a spectrogram using this lower frequency limit, obtained by the EB040 observatory on 2025-11-12. This day was affected by a severe geomagnetic storm, causing depletion of the ionosphere and thus limiting oblique propagation of higher frequencies. However, no flares above class C5 occurred on this day. A few additional broadcasting bands can be clearly identified in this figure between 2.5 and 10 MHz, as horizontal features in the spectrogram as well as in the time integrated intensity shown to the right of the spectrogram. The diminishing of reception of the lower frequencies around the middle of the day is even more pronounced at the frequencies below 10 MHz, as is to be expected. The broadcasting bands observed below 10 MHz might be

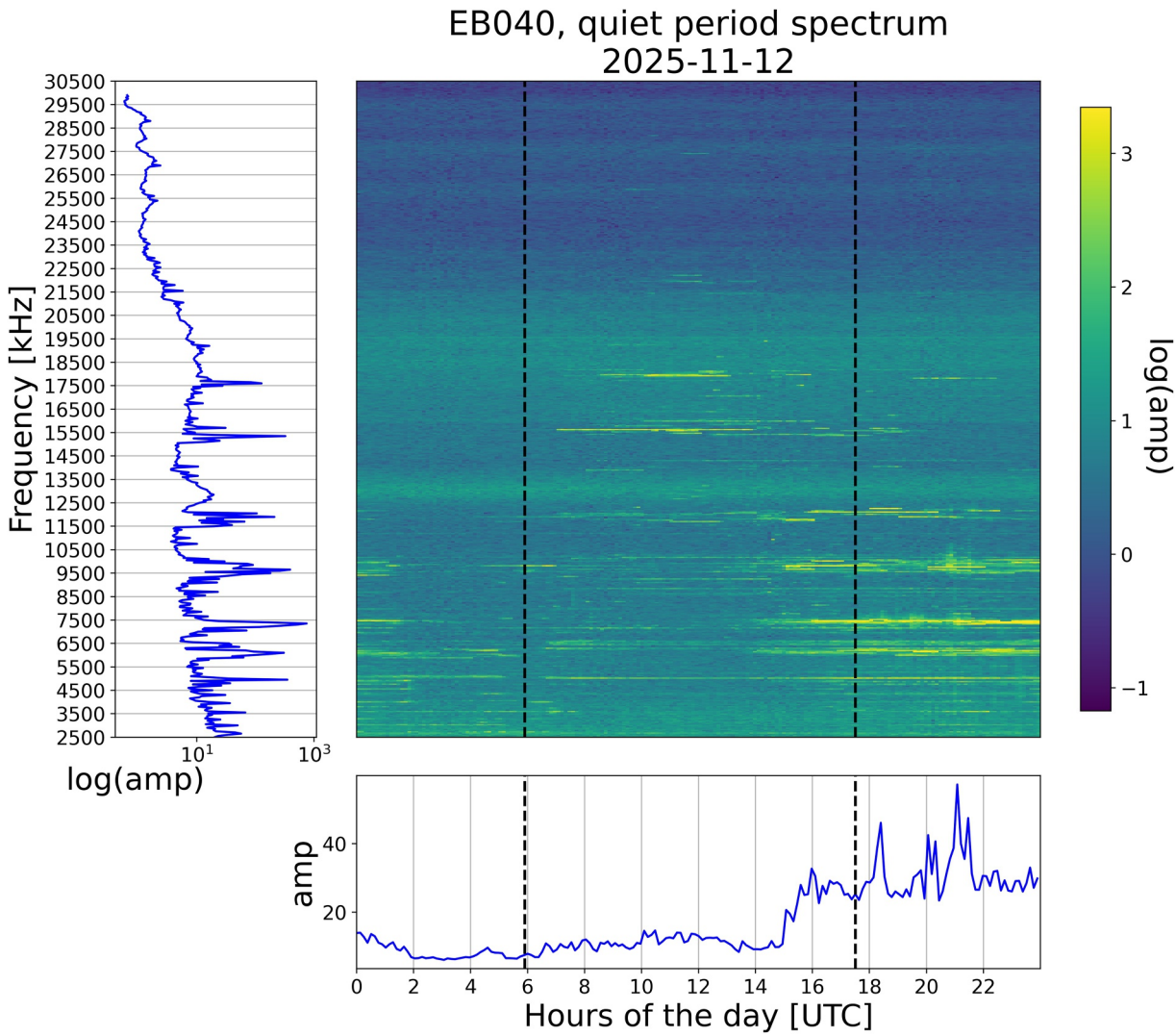


Figure 4. Spectrogram obtained by the EB040 ionosonde on 2025-11-12 (a quiet day, no flare occurred above class C5) showing frequencies from 2.5 to 30 MHz. Apart from the lower frequency limit, the presentation is similar to that of Figure 3.

especially suitable for monitoring the impact of particle induced absorption at night. However, in the remainder of the current work we focus on the frequencies above 10 MHz because measurements at lower frequencies have only been started recently and only at a few observatories, so there is not yet a significant amount of data available.

3.3. Observations in Selected Frequency Bands

As seen from the quiet spectrograms, there are various broadcasting bands in which a clear signal, propagating via sky-waves, can be detected. For the purpose of monitoring ionospheric *D*-layer variations, these bands are more useful than the overall spectrum, as the power law part of the recorded spectrum is likely due to local emissions or instrumental noise and contains no information about ionospheric propagation conditions. This part of the spectrum should also contain the extraterrestrial emissions which are exploited in RIOMETER observations and which are affected by the ionisation level in the *D* region. Since this portion of the received spectrum does not appear to exhibit variations that can be attributed to ionospheric variation, we conclude that our technique is not able to clearly identify extraterrestrial contributions to the total received signal. Similarly, there seems to be little variation observable in the broad band atmospheric noise produced by lightning strikes which is a source of noise in SuperDARN observations (Berngardt et al., 2018; Bland et al., 2018). Therefore, as a next step, the selected

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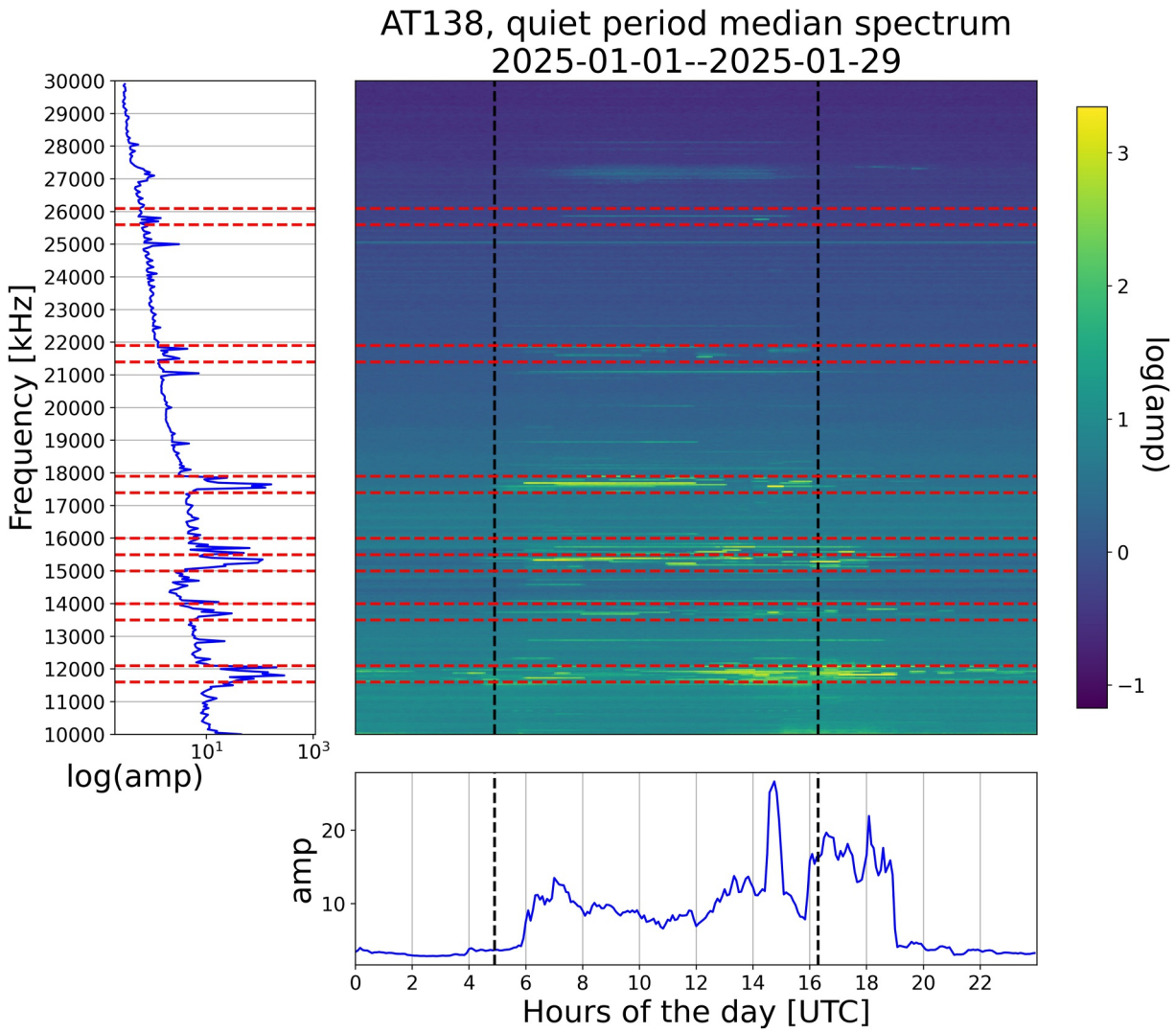


Figure 5. Median spectrogram from AT138, covering 1–29 January 2025, with the frequency integrated time-series below the spectrogram calculated by summing only over the broadcasting bands on the spectrogram marked by horizontal red lines. The times of sunrise and sunset are again indicated with vertical lines.

bands in which enhanced anthropogenic emissions are observed via sky-wave propagation are analysed separately.

Figure 5 shows a median spectrogram obtained from AT138 for the period from 1–29 January 2025, similar to panel D of Figure 3. However, the diurnal time series shown in the panel below the spectrogram is calculated here by summing only those frequencies within the bands marked by red, horizontal, dashed lines. These bands are each 500 kHz wide, and centred at 11.85, 13.75, 15.25, 15.75, 17.65, 21.65, and 25.85 MHz. These bands are chosen to have all identical width and cover the broadcasting bands mentioned above (the different broadcasting bands themselves have varying widths). It is evident from the figure that most of the diurnally varying features observed are indeed seen in these selected bands. The qualitative features in the frequency-integrated time series are therefore very similar to those seen in Figure 3.

4. Effects of *D*-Layer Enhancements

Here we give some examples of the effects of *D*-layer enhancements as seen in the radio-quiet spectrograms recorded by ionosondes. In the current work, the intend is to demonstrate the possibility of detecting HF absorption events using these data. More qualitative analyses will be presented in subsequent work. We consider *D*-layer ionisation produced either by solar flare X-rays or by particle precipitation.

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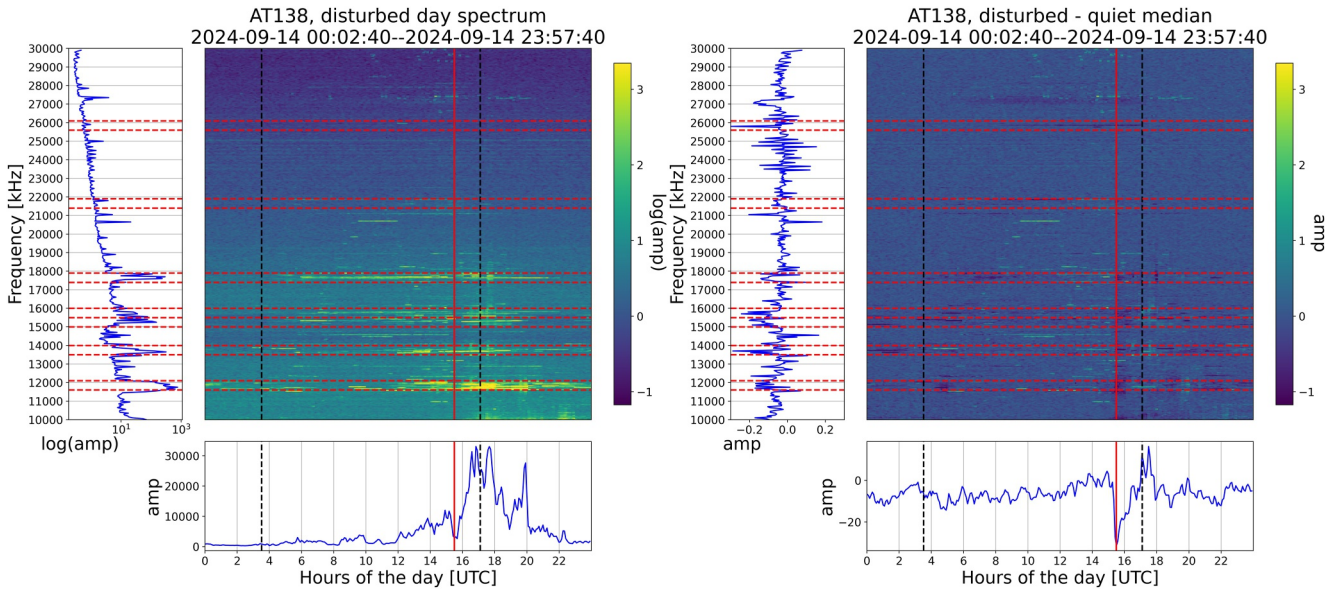


Figure 6. Spectrogram of the day (left) and relative deviation from the median (right) observed by the AT138 ionosonde on 2024-09-14. The time of the X4.5 class solar flare is indicated with a red vertical line.

4.1. Solar Flare Effect

Figure 6 shows an example of the effect of an X4.5 class solar flare that occurred on 2024-09-14, reaching its maximum at 15:29 UTC (the time of the flare maximum is indicated by a red line on the figure). The left hand panel shows the spectrogram recorded on this day by the ionosonde AT138. To the left of the spectrogram is the time integrated spectrum while below the spectrogram is shown the frequency integrated time series, considering only the seven broadcasting bands (marked by horizontal lines) that were identified as showing the most sky-wave noise. The typical diurnal patterns described before can be seen, but also a clear depletion in the received amplitudes just after the occurrence of the flare. This is more obvious in the time series integrated over the selected frequency bands than on the spectrogram itself.

The right side panel of Figure 6 shows the residuals obtained by subtracting the quiet-time median for September 2024 from the spectrogram shown on the right. In this case, the time integrated spectrum shown to the left of the residual spectrogram of course only shows some noise (with small amplitude) around zero. The frequency integrated time series below the spectrogram shows slightly negative values throughout the day. This indicates that the X-ray flux—and therefore also the *D*-layer ionisation and the level of absorption—were somewhat higher than the median on this day. Indeed, for most of this day, the solar X-ray flux was above the level of a C3 class flare. The most interesting feature, however, is the very clear, sharp drop in residual at the onset of the flare, followed by a typical recovery profile.

Figure 7 shows the time series integrated over the range from 15 to 16 MHz (this combines two of the selected bands which are adjacent to each other) as measured by the DB049 ionosonde on 3 October 2024. This was a geomagnetically unperturbed day. The vertical pink line marks the peak at 12:18 UT of an X9-class solar flare. From the spectrogram (e.g., the left side panel of Figure 2) it can be seen that this band normally shows strong reception of broadcasting transmissions during the day time, so Figure 7 covers the range from 08:00 to 16:00 UTC. The sharp increase seen at 16:00 UTC is the beginning of the greyline propagation enhancement around sunset.

The blue line and shaded area show the median and interquartile range of the amplitude integrated over the frequency range from 15 to 16 MHz, calculated over a period including 14 days around the day of the flare. The red dots show the integrated amplitude observed on the day of the flare. A clear decrease of the integrated amplitudes can be seen, with measured values well below the lower quartile for 1 hour and reaching its minimum around the time of the peak of the flare.

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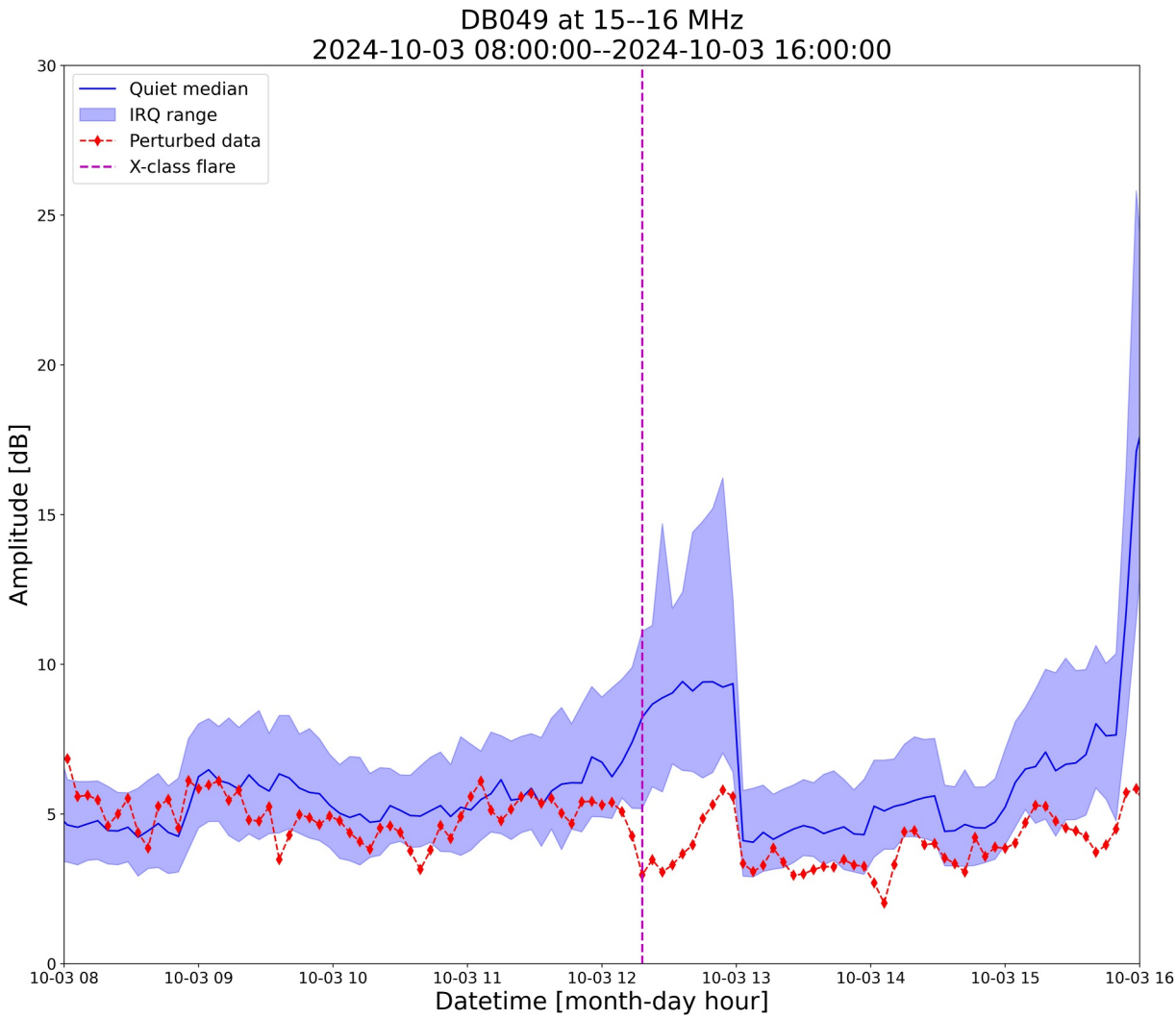


Figure 7. Example of the effect of an X9-class solar flare on 2024-10-03. The blue line shows the median, while the shaded area indicates the interquartile range. The red dots indicate the measured noise integrated between 15 and 16 MHz, and the peak of the flare is marked by a vertical dashed line.

Figure 8 Shows the diurnal spectrogram from DB049 for 2025-11-11. On this day, a solar flare of class X5.1 occurred, reaching its maximum flux at 10:04 UTC. This spectrogram includes also the frequencies between 2.5 and 10 MHz. Long distance signals below about 5 MHz are not detected around the time of this flare, due to the presence of *D* and *E* layers as described above. The various frequency bands seen between about 5 and 22 MHz again all show a very clear, abrupt decrease at the time of the flare. The time-series below the spectrogram shows again the sum of intensities in the seven selected bands discussed above, although the bands are not explicitly indicated on the spectrogram. This integrated amplitude decreases to almost zero during the flare.

Interestingly, above about 22 MHz, a small increase in the signal can be seen for frequencies where no broadcasting band is otherwise detected. This represents the lowest frequencies detectable on the ground of a solar radio burst that accompanied the flare. This may indicate the potential use this method to detect solar radio bursts. The frequency integrated time series below the spectrogram shows a sharp drop very close at the time of the solar flare maximum flux followed by a recovery to the trend values after this time. The latter is also indication of enhanced *D*-layer ionisation and enhanced level of absorption for the given radio-frequency range.

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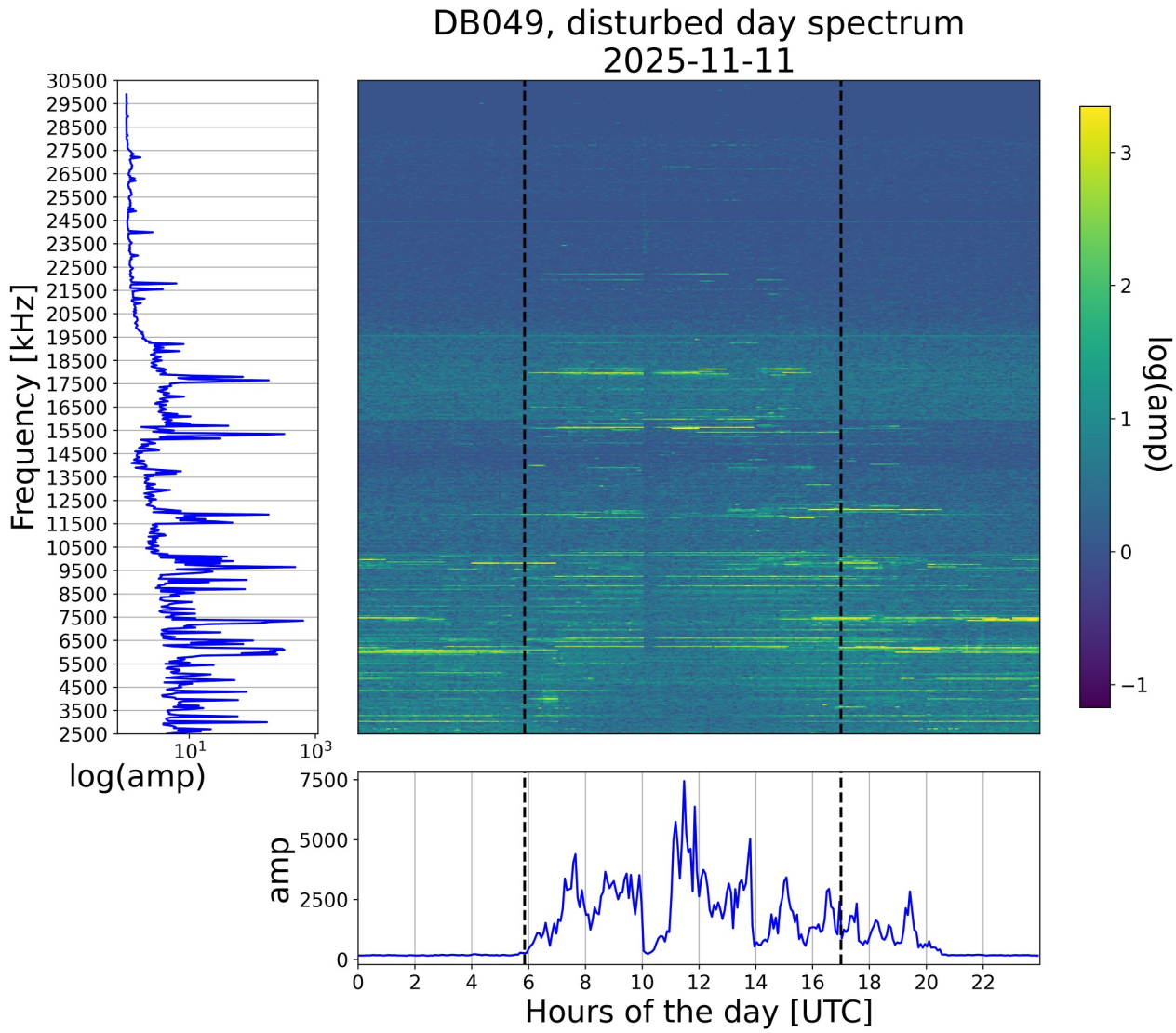


Figure 8. Spectrogram obtained by the DB049 observatory on 2025-11-11, showing the effect of an X5.1 flare at 10:04 UTC (the time of the flare is not marked here; black lines mark the times of sunrise and sunset).

4.2. Particle-Induced Absorption

In Figure 9 we show an example of an increase in absorption due to the presence of a particle-induced *D*-layer. This Figure shows the residuals obtained by subtracting the median spectrogram from the spectrogram recorded by AT138 on 10 May 2024. Most of the seven days preceding and succeeding this event were not affected by long-lasting, particle-induced absorption enhancements, so we consider the medians calculated from these days to adequately represent the quiet conditions. On this day, a severe geomagnetic storm started around 18 UTC, with a moderate solar particle storm happening at the same time. Earlier on the same day, solar flares of class X4 (at 06:54 UTC) and M6 (at 14:11 UTC) occurred, readily allowing comparing the signatures of both types of events. The absorption due to the particle event can be seen to be more severe than those due to the solar flares. This is particularly evident when looking at the frequency integrated time series shown below the spectrogram.

It can be seen that in this case even at the AT138 ionosonde, which is located at 38°N (see Table 1), the particle event has a more significant impact than the solar flares. It can also be seen that the particle-induced *D*-layer enhancement lasts much longer than those associated with the flares. For this particular event, the particle event lasted until the next day, when the situation became more complicated due to further high-energy particle events (Papaioannou et al., 2025). In general, particle-induced D-layers can form due to either solar proton events or

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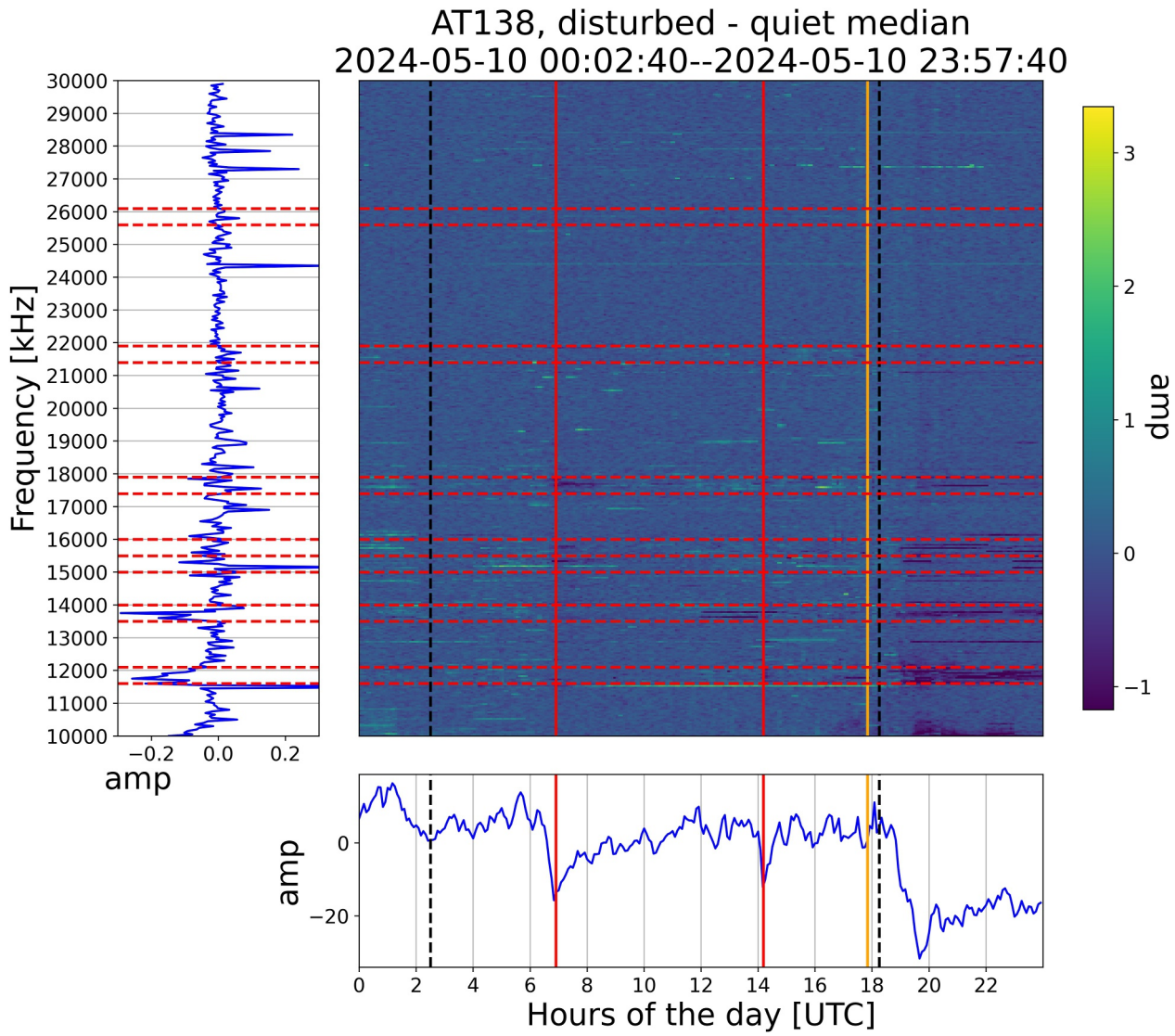


Figure 9. Spectrogram of the relative deviation from the median observed by the AT138 ionosonde on 2024-05-10. The time of the X4 and M6 class solar flares are indicated with a red vertical line, while the orange line shows the peak time of the solar proton event.

precipitating particles in the auroral oval during a geomagnetic storm. During the complex event of 10 May 2024 both phenomena occurred (Fiori et al., 2025), but the latitude of the AT138 observatory and the timing of the absorption event indicate that Figure 9 shows the impact of an auroral absorption event. Further analyses of this event are out of the scope of this paper and might the subject of future research.

5. Conclusions and Future Work

We analysed for the first time the radio quiet spectrograms recorded by various European ionosondes, showing clear signatures of broadcasting signals in the HF band. Since these signals are received via sky-wave propagation, they can be exploited as transmitters of opportunity for monitoring the level of absorption at various frequencies throughout the HF band. This type of measurement does not require a lot of time in the ionosonde observations schedule and can be implemented at existing observatories without the need to deploy any new instruments or antennas. Thus, it provides a relatively easy way to improve the real-time monitoring of the level of HF absorption in the lower ionosphere.

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We showed some examples of enhanced radio absorption in the ionosphere resulting from solar flares or particle precipitation events. From these, it is evident that the absorption events have a signature in the data that is, at least qualitatively, easy to identify. Evidently, for X-class flares, the signature is unmistakable and it should allow for automated detection. This would provide a way of detecting absorption events directly in the ionosphere, without relying on satellite observations of X-ray or particle fluxes.

The current work only serves to demonstrate the proof-of-principle in a qualitative way, without yet quantitatively relating the strength of flares and particle events to the level of absorption in different bands. Especially for solar flares, for which many examples are present even in the few years of data available, we plan to expand the work to such a quantitative analysis. For particle events this may prove more difficult, as there are fewer examples and their impact might be more complicated because it is also influenced by the geomagnetic field conditions.

The measurements analysed here are of radio signals propagating via ionospheric reflection, as opposed to trans-ionospheric radio waves used in RIOMETERS. This presents a complication, because the received amplitude at any given time for a particular frequency will depend not only on the state of the *D* layer, but also on the conditions of the *E* and *F* layers responsible for the reflection of the radio transmissions. The technique presented here is therefore more similar to the monitoring of ionospheric radio absorption in SuperDARN radar data (Berngardt et al., 2018; Bland et al., 2018, 2022). The effects of the typical diurnal pattern in the *E* and *F* layers can be readily seen in the quiet-time spectrograms such as those shown in Figure 2: different frequency bands are received more or less at certain times of day, depending on availability of long-distance propagation channels via the reflecting layers of the ionosphere. In order to detect, and possibly quantify, the effect of solar flares or particle-induced absorption events it is therefore necessary to compare the observed noise levels to a median diurnal curve, which we calculate for each day by considering the seven preceding and succeeding days. Since different frequencies will be able to propagate more or less under different conditions of the *E* and *F* layers, the medians need to be calculated separately for each frequency band.

The example of a major solar flare discussed in Section 4.1 and illustrated in Figure 7 suggests that effects of absorption events can be detected based on a decrease in the received amplitude from the median below a set threshold, for example, based on the interquartile range. Such a quantitative detection level would be needed in any type of future operational exploitation of the type of observations presented here, and determining the appropriate thresholds to be used will be part of a more quantitative extension of the current work, which should provide a validated indicator for robust detection of effects of absorption events.

While our main interest is the detection and characterisation of radio absorption events, the various features seen in the median spectrograms could be of interest for further investigations, precisely because they contain information about the propagation conditions provided by the *E* and *F* layers. For instance, the multi-peaked pattern seen during the greyline propagation period for most seasons (Figure 3) warrants further investigation. A better understanding of the quiet-time patterns present in the spectrograms may in turn aid in improving the construction of the quiet diurnal curve used to detect and characterise absorption events.

In future work, the effect of solar flares in each of the HF broadcasting bands will also be investigated separately, instead of summing all bands together, as a function of the strength of the flares. This will allow establishing a relation between flare class and the observed absorption as a function of the frequency throughout the HF radio band. This would provide a direct assessment of the level of absorption at various frequencies without the need for a model to relate for example, X-ray flux to strength of absorption. The obtained results can complement observations by spectral RIOMETERS, which can perform similar measurements, including at slightly higher frequencies (typically up to the lower VHF band).

It might also be of interest to reconsider the configuration parameters of the ionosonde used to make radio-silent spectrograms. In particular, since integrating only over selected frequency bands turns out to provide the best opportunity for detecting absorption events, a measurement program could be set up recording only those frequencies. This could allow for longer integration times for the frequencies of interest, or for shortening the measurement program (which in turn might allow running it at a higher time-resolution).

Appendix A: Configuration of the Instrument

The data collected for the current work was obtained using Digisonde DPS4D ionosondes. Each instrument employed an identical measurement sounding program for this purpose. The exact settings used are given below, allowing any DPS4D operator to perform similar observations.

Frequency stepping	
Coarse Freq Law:	Linear
Lower Freq Limit:	10,000 kHz
Upper Freq Limit:	30,000 kHz
Coarse Freq Step:	25 kHz
Number of Fine Steps	none

Note. For some measurements, starting in October 2025, the lower frequency was set to 2,500 kHz instead.

Range sampling	
Start Range:	0 km
Number of Samples	256
Inter-Pulse Period	auto

Pulse integration	
Number of Integrated Repeats	1
Interpulse Phase Switching:	disabled

Note. Since only 1 pulse is used, the interpulse phase switching setting is irrelevant.

The gain for the receiver is set to the maximum, O-polarization is selected only, and the system is set to radio silent operations. No radio frequency mitigation techniques are employed, and an average over time is calculated and saved as a TAV file.

Appendix B: Accessing TAV Files From GIRO

The radio quiet spectrograms measured by various ionosondes are freely available from the GIRO data repository (Reinisch & Galkin, 2011), including the various examples used in this article. The URL at which data can be accessed is https://lgdc.uml.edu/digilisten/getscans?ursiCode=<URSIcode>&fromDate=<fromDate>&toDate=<toDate>, where <URSIcode> is the URSI code of the ionosonde, and <fromDate> and <toDate> are the begining and end of the requested period (the maximum length for a requested period is 24 hr). For example, the request https://lgdc.uml.edu/digilisten/getscans?ursiCode=DB049&fromDate=2024.05.10T00:00&toDate=2024.05.10T23:59 will return de data for the entire day of 2024-05-10 for the Dourbes ionsonde (DB049).

The data will be delivered in a plain text file, with a straightforward format. At the top of the file is a block of information regarding data usage and licensing. Following this are the spectrogram records, in chronological order. Each spectrogram record begins with a small metadata block, followed by the actual data. An example of the beginning of a spectrogram record is shown in Figure B1.

Each spectrogram record begins with a short metadata block identifying the observatory and giving the co-ordinates and the time of the measurement. The first column of data gives the frequency (one line for each frequency that was sampled), and the following columns list the amplitudes and phases recorded by each of the four receiver antennas connected to the instrument. Normally, the differences between amplitudes recorded by the four antennas should be small, unless a local source of interference is affecting some antennas significantly more than others. The phases of the received signals are not used in this work.

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```
#
# Location: GEO ( 50.1 N     4.6 E ), URSI-Code DB049 (Dourbes)
# START UT: 2024-05-10T00:27:00.000Z
#
# Freq,kHz          Antenna 1           Antenna 2           Antenna 3           Antenna 4
#            Amp,lin Phase,deg   Amp,lin Phase,deg   Amp,lin Phase,deg   Amp,lin Phase,deg
 10000.000   25.028    70.313   17.670    15.469   42.189   147.656    8.808    71.719
 10025.000    4.025   176.484    2.189   125.859    2.841   279.141    5.225   121.641
 10050.000   10.483    28.828    8.808    31.641    2.605   350.156    6.219    52.031
 10075.000    3.100   275.625    4.790   356.484    2.006   160.313    2.006   351.562
 10100.000    6.784    26.719    3.100    19.688   12.476   317.813   14.848    25.312
 10125.000    8.808   159.609    2.006   288.984    4.391    92.813    4.391   191.250
 10150.000   11.436    26.016    8.074   329.063   13.610    82.969   12.476   109.687
 10175.000    7.401   340.313    3.689   261.562    6.219   104.063    1.545   324.141
 10200.000    5.225    84.375    2.841   335.391    7.401   326.250   14.848   144.141
 10225.000    4.025   217.969    2.189    21.094    4.025   310.078    9.609   206.719
 10250.000    1.000   184.922    2.841   194.063    8.808   168.750    3.382   327.656
 10275.000    4.790   358.594    3.100   277.734    5.225   234.844    4.391   314.297
 10300.000   12.476   127.969   10.483   138.516   14.848    90.703    9.609   108.984
 10325.000   21.030   179.297   13.610    33.047   17.670     7.734    5.225   168.047
 10350.000    2.841   297.422    3.689   166.641    6.219   249.609    1.545   334.688
 10375.000    5.700   321.328    1.839   189.844    4.025    67.500    6.219    35.156
```

Figure B1. Example of the beginning of a spectrogram record obtained from the GIRO data repository. As shown in the metadata block, this observation was obtained by the DB049 ionosonde on 2024-05-10 at 00:27 UTC. Note that this record is from night-time for frequencies starting at 10 MHz, so no strong signals are present in this example.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The radio-quiet observations made by ionosondes are freely available from the GIRO repository (Reinisch & Galkin, 2011). Details on how to access this data and a description of the data format are provided in Appendix B of the current paper. The code used to produce the various spectrograms and time-series shown in this paper is available in the zenodo repository Buzás (2026).

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